

# Advances in science

Great scientists are thinkers who understand the world around us, provide solutions to problems, and create new things. This has led to many great inventions and discoveries.

## Johannes Gutenberg (c.1398–1468)

Gutenberg played a key role in printing. Experts believe he invented metal-type printing in Europe. Gutenberg's press was quick, accurate, and durable, compared to earlier woodblock printing.



Gutenberg's first printed book was the Bible in 1455.



A falling apple probably inspired Newton to think about gravity.



In a rainbow, white light breaks up into seven colors.

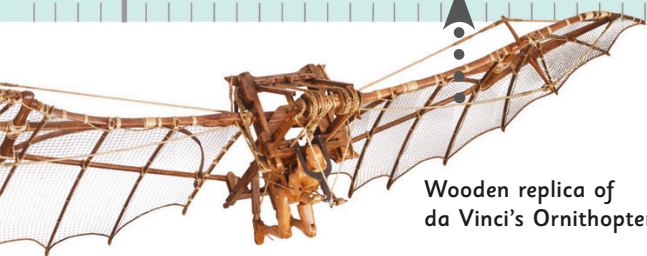
## Isaac Newton (1642–1727)

Newton investigated forces and light. He realized there must be a force that keeps the planets in orbit around the Sun. This force is known as gravity. Newton also discovered that white light is a mixture of lots of different colors.

1400

1500

1600



Wooden replica of da Vinci's Ornithopter

## Leonardo da Vinci (1452–1519)

Da Vinci was a painter and inventor. He drew plans for helicopters, airplanes, and parachutes. Unfortunately, the technology of the time was not good enough to build a working model for any of these.

## Galileo Galilei (1564–1642)

Galileo proved that the Earth moves around the Sun by looking at the solar system through a telescope. A few wise thinkers had always suspected the truth, but most people at the time believed that our Earth was the center of everything.



Replica of a 17th-century telescope